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Healthcare Analytics: Examining the Diagnosis-Treatment Cycle

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Abstract

Research has demonstrated that the patients' diagnosis-treatment cycles often significantly deviate from the standardized clinical pathways. Analyzing these deviations might result in the further enhancement of the quality of care, the promotion of patient safety, an increase in patient satisfaction and an optimization of the use of resources. Understanding pathway behavior and deviations becomes possible because of an increased availability of reliable data, which originates from the hospitals information systems.

In this paper we propose a clinical pathway analysis method for extracting valuable medical and organizational information on past diagnosis-treatment cycles that can be attributed to a specific clinical pathway. The method is applied on the clinical pathway processes in a Gynecologic Oncology Department.

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1. Introduction

Clinical pathways represent prescriptive models of the standard healthcare procedures that need to be undertaken for a specific patient population [1]. Instances of the clinical pathways (also known as cases) describe the actual diagnostic-therapeutic cycle of an individual patient. Typically real-life clinical pathways are characterized by the following unique set of properties:

- **High flexibility.** Patients most often do not come with the exact same mix of medical conditions (e.g. heart conditions, pregnancy, allergies, etc.) [2]. Additionally, medical experts have to deal with uncertainty during the treatment, e.g. unexpected outcomes such as adverse drug reactions, drug allergies, heart failure or other complications.
- **Complex decision making.** While the medical knowledge used in the decision process comes partially from published research contributions and widespread medical guidelines (with various kinds of evidence levels), it is generally accepted that the decision process is profoundly influenced by the expertise and experiences of the involved medical experts. High levels of autonomy are commonly awarded to the experts, which results in a high responsiveness in critical situations [3].
- **Network of specialists.** While a horizontal job specialization allows those experts to acquire a deep knowledge within a specific medical discipline, it often renders it impossible for an expert to execute every activity in a clinical pathway. Consequently, clinical pathways are predominantly performed by a network of collaborating medical specialists, all with high levels of autonomy to decide on their working procedures [4]. Clinical pathways enable the coordination between medical experts and their specific activities.
- **High information needs.** Within clinical pathways there exist high information requirements both in terms of accuracy and availability. Patient specific and situational information are crucial for the decision making processes of the individual medical experts. Consequently, a large variety and amount of the patient specific information must be exchanged between actors both within a pathway instance and between different pathway instances [5]. As a patient can be involved in multiple pathway types, inter-pathway information exchanges may take place between pathways of different types.
- **Continuous evolution.** A strong academic background, experience-based insights and technological developments as well as economic and governmental pressures, contribute to an often implicit but continuous evolution of the medical practices and the related clinical pathways [6].

Medical informatics and decision making researchers have embraced the concept of clinical pathways and have used these pathways for configuring a wide spectrum of healthcare information systems, e.g. in [7-13]. The main objective of this approach has been the enhancement of the quality of care by improving risk-adjusted patient outcomes, promoting patient safety, increasing patient satisfaction and optimizing the use of resources [14,15].

In reality, the care processes of individual patients deviate from the standardized path in order to accommodate requirements dictated by the patient's specific medical conditions or the local conditions of the healthcare organization [16,17]. This study presents a novel approach that is based on process mining techniques to acquire insight in the real sequence of healthcare activities performed on specific patients. Additionally, the study reports on the results of a case study in a Gynecologic Oncology department.

2. Methods

2.1. Subjects and Data Source

The study has been executed on the care processes of 1143 patients of the Gynecologic Oncology department of the a major European academic hospital. This data has been made available in the context of the Business Process Intelligence Challenge [†], for which we obtained an award for most the complete analysis. All diagnosis and treatment activities that were performed for these patients during a period of 3 years, from January 2005 to January 2008 were collected. The selected patients were all diagnosed with a cancer pertaining either to the cervix, the vulva, the uterus or the ovaries. In total 236 treatment combinations are covered by this set of patients. Different patient subsets were retrieved for examining both radiotherapy and chemotherapy cancer treatment.

A myriad of information has been recorded for each individual activity including the patient identifier, the activity type, the timestamp, the performer's department indicators and information on both the diagnosis and treatment type. Exactly 150,291 activities or an average of 131.5 activities per patient (spread: 1-1806 activities per patient) were recorded. All activity types (677 in total) are fine-grained and directly related to diagnostic and therapeutic activities of the gynecologic oncology pathways on which the core processes are based. We filtered out all administrative related activities, such as 'order rate' (translation of the activity 'ordertarief'). An example extract of the data set can be found in Table 1.

Table 1: Activity log extract

Patient	Event type	Treat.	Dep.	Diagnosis	Time
155	follow-up polyclinic consultation	61	SGNA	gyn. tumors	1-Jan-05
156	cytological examination vagina	61	LVPT	gyn. tumors	1-Jan-05
156	histological examination	61	LVPT	gyn. tumors	1-Jan-05
275	teletherapy	13	RATH	gyn. tumors	1-Jan-05
275	follow-up polyclinic consultation	13	SGNA	gyn. tumors	1-Jan-05
336	potassium flame photometry	603	CHE2	malign cervix	1-Jan-05
336	differential count	603	HAEM	malign cervix	1-Jan-05
336	determination trombocyte level	603	HAEM	malign cervix	1-Jan-05
336	count of leukocytes	603	HAEM	malign cervix	1-Jan-05
10	count of leukocytes	113	HAEM	malign cervix	4-Jan-05
10	determination trombocyte level	113	HAEM	malign cervix	4-Jan-05
72	differential count	3101	HAEM	malign ovary	16-Jan-05

2.2. Clinical Pathway Analysis Method

In this section we introduce the three broad application areas for process mining techniques in a healthcare setting: analyzing recurring patterns, pathway variants and exceptional/adverse events (see Figure 1).

Discovering recurring patterns. Highly specialized medical experts are often confronted with 'similar' instances of this cause-effects-solutions relation, as a consequence they might unconsciously develop optimizations of the clinical pathways. Process discovery and visualization techniques summarize full and precise insights on the clinical pathways reality in one visual. Process discovery techniques, such as the heuristics miner [18], can be used to obtain models of the as-is pathways.

[†] Hospital log, TU Eindhoven. <http://dx.doi.org/10.4121/uuid:d9769f3d-0ab0-4fb8-803b-0d1120ffcf54>

Analyzing and characterizing pathway variants. The medical condition of patients is determined by a broad set of characteristics/factors, e.g. type and location of the tumor, age of the patient, pregnancy, allergic reactions, etc. By comparing the specific medical conditions of patients with the factors used for characterizing the different clinical pathway variants, the medical expert can identify the best therapeutic option. Different approaches can be used for identifying new clinical pathway variants. Firstly, the analyst could perform a correlation analysis on the factors, which are usually hidden in the data perspective. After the uncovering of important factor correlations, traces can be filtered and the common process characteristics for this process variant can be documented. Secondly, a trace clustering technique can be used to cluster pathway traces (i.e. a form of unsupervised learning), followed by an analysis of the factors in each cluster [19]. Finally, an expert can propose interesting factor combinations (e.g. combination of treatment & diagnosis code).

Identifying and analyzing exceptional medical cases and/or adverse medical events. Two broad sets of techniques are of main interest for uncovering both exceptional cases and adverse events: detecting inconsistencies between a designed prescriptive model and the real-life process (i.e. conformance checking [20]) and verification of specific process properties (i.e. rule-based property verification techniques [21]).

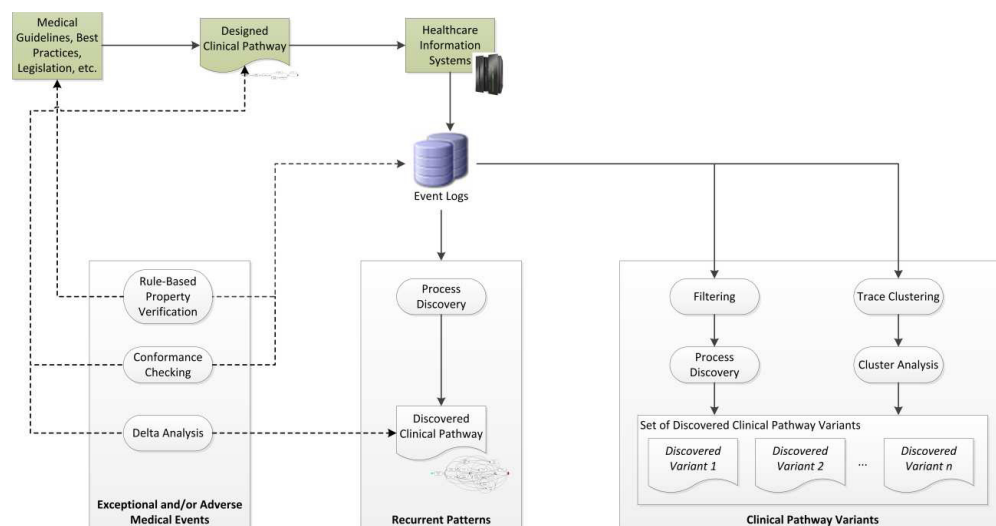


Figure 1: Clinical Pathway Analysis Architecture

3. Results

3.1. Clinical Pathway Discovery: Recurrent Patterns in Radiotherapy-Based Ovarian Cancer Treatment

Figure 2 depicts the retrieved care process model of the activities performed by the radiotherapy department. This process model consists of three clearly distinguishable blocks: (1) the medical consultations, (2) the preparatory activities – i.e. the “simulation” activity and the “treatment time” calculation activity – and (3) the radiotherapy activities themselves: teletherapy, hyperthermia therapy and brachytherapy.

The heuristics miner [18] with standard parameterization has been used for the construction of this care process model. Models obtained with the heuristics miner typically represent general behavior, as it uses frequency thresholds for the selection of the discovered activity ordering relations that will be depicted. The numbers on the arcs (representing individual activity sequences) in the process model indicate respectively the

frequency- and counterevidence- based level of certainty of the existence of that activity sequence relationship (between 0 and 1) and the number of times that sequence between the activities has taken place.

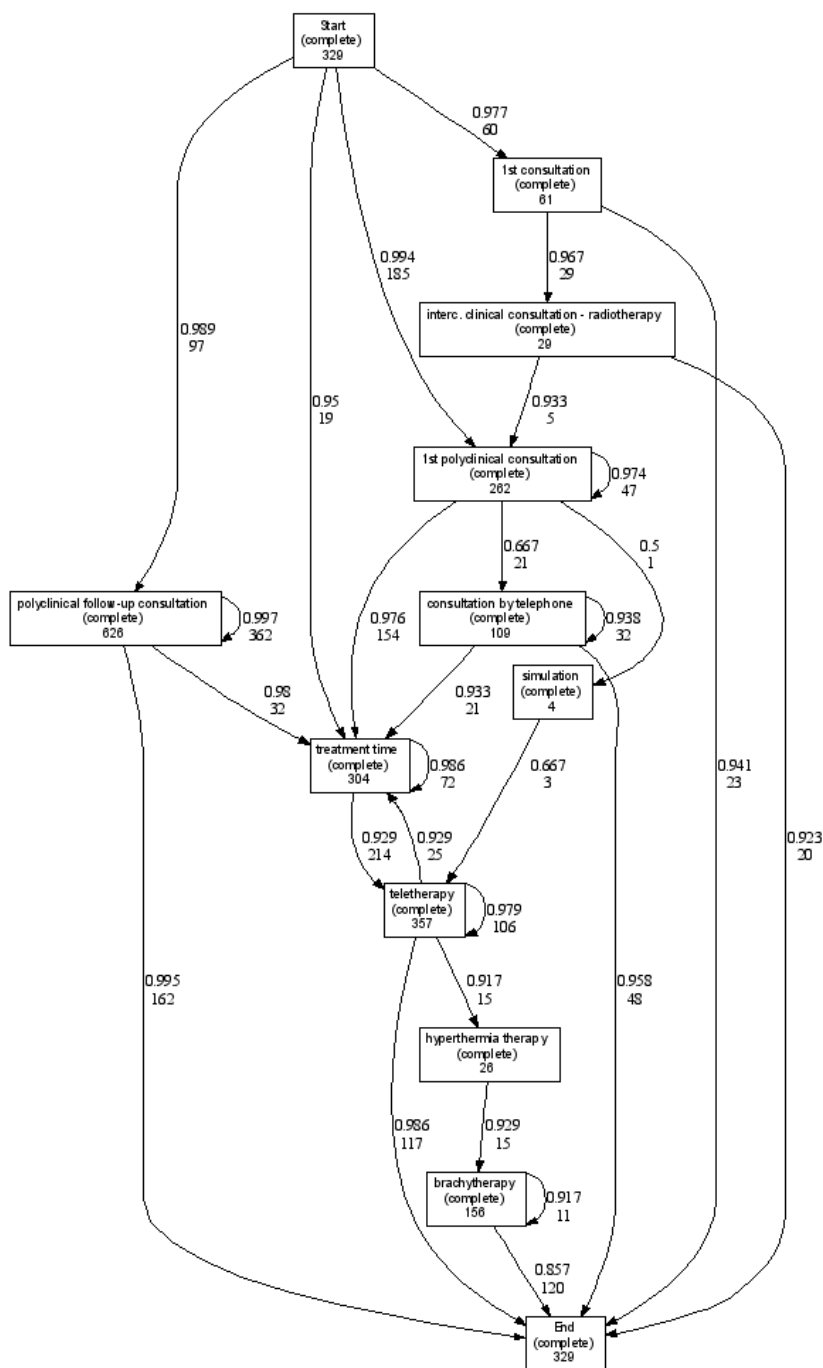


Figure 2: Recurrent patterns in radiotherapy-based ovarian cancer treatment

3.2. *Exceptional Medical Events: Deviations from the Common Radiotherapy Activity Sequence*

In this section we further investigate the common treatment sequence – teletherapy, hyperthermia therapy and brachytherapy – that has been suggested in the retrieved care process model.

Of the 215 care process instances containing at least one radiotherapy treatment, 190 instances (89%) start with teletherapy. Hyperthermia only occurs in a minority, 22 cases or approximately 10%, of the care process instances. Extensive testing of these 22 cases revealed that in approximately 27% the hyperthermia therapy is followed by an additional teletherapy treatment. In general, we conclude that teletherapy and hyperthermia are coupled and that teletherapy precedes hyperthermia. In 144 pathway instances the brachytherapy occurs at the end of the care process. In 83% of the cases including brachytherapy, the brachytherapy occurs directly after the teletherapy. Approximately 93% of the cases that include hyperthermia end with brachytherapy.

The adherence to this treatment sequence has been assessed with the LTL-checker tool [21]. LTL-checker allows for testing the activity log against (un)expected/(un)desirable process properties that are specified in temporal logic.

3.3. *Variance in Paclitaxel-Chemotherapy-Based Treatment*

In this section, we summarize the key observations on the use of Paclitaxel for ovarian cancer over the different stages. The ovarian cancer disease stage has a significant influence on the recommended chemotherapy [22], and thus results in different variants of the care flow. Visualizations of the variants similar to the one provided in Figure 2 can be obtained but were left out due to space restrictions.

Ovarian Cancer Stage Ia. In [23] it is concluded that patients diagnosed with ovarian cancer stage Ia do not require postoperative chemotherapy based on the active ingredient paclitaxel. Accordingly, the three patients (patient 1049, 1067 and 1139) diagnosed with stage Ia ovarian cancer did not receive this type of chemotherapy. However, we only found activities performed by the operating room in the care path of patient 1049.

Ovarian Cancer Stage Ic and II. Further postoperative chemotherapy is recommended for patients with ovarian cancer stage Ic or stage II [24]. However, out of the five patients with this medical condition, only three have undergone surgery (patient 819, 987, 1020). But none of them received a postoperative paclitaxel treatment of at least three cycles, as described in [25].

Ovarian Cancer Stage III. In their literature review Herzog and Herrin [22] list three standard (after surgery) chemotherapy courses for stage III ovarian cancer. Each of these courses covers six 21-day-cycle treatments based on a combination of paclitaxel with carboplatin or cisplatin. While the majority of the paclitaxel administration schedules start with three 21-day-cycles, only the schedule for patient 829 comes close to the six 21-day cycle treatment. Based on the activity log we were unable to conclude that any patient received the combination of paclitaxel and carboplatin or cisplatin.

AC-Paclitaxel Chemotherapy. Several sources also indicated the benefits of the sequential use of AC (i.e. doxorubicin and cyclophosphamide) and Paclitaxel, both in terms of chances for disease-free survival and for overall survival (e.g. [10,26]). However, we were unable to find any patient who receives both the AC and Paclitaxel therapy. A minority of the patients received a combination of Paclitaxel and Doxorubicin (indicated as ‘doxorubicine liposomal –caelyx’ in the event log).

4. Discussion

4.1. Radiotherapy for Ovarian Cancer

Based on the retrieved care process model for the radiotherapy department we were able to uncover different types of rather exceptional behavior. Firstly, this care process model indicates that a high level of the patients in this workflow never receive a radiotherapy treatment, i.e. transitions between a consultation related activity and the end state.

Secondly, under extremely exceptional circumstances, only 4 times according to the event log, a simulation activity has been explicitly performed. As a simulation is crucial and the non-execution of this type of activity could result in major quality risks, we assume that the registration of this activity has not been done systematically or that it is contained in the “treatment time” determination activity.

Thirdly, while we established a clear sequence between the treatment activities, several patients were treated in a different way. For example, multiple patients diagnosed with cancer pertaining the cervix that received radiotherapy treatment (indicated by the presence of teletherapy), did not receive brachytherapy (e.g. patients 447, 451 and 457). However, according to Gerbaulet et al. brachytherapy is a standard of care for cervical cancer [27]. Many plausible explanations may exist, such as the patient’s decision to stop the cancer treatment or the performance of the brachytherapy at a different hospital.

4.2. Chemotherapy for Ovarian Cancer

Two major deviations from medical guidelines were uncovered during the analysis of the administration of paclitaxel-based chemotherapy: the lack of combining platinum agents with paclitaxel and the lack of a postoperative paclitaxel-based chemotherapy (and of the surgery itself).

Secondly, gynecologic oncology research has extensively studied the optimal composition of chemotherapy regimens for the treatment of ovarian cancer, both for the first- and second-line treatment. While there has been no general agreement on the optimal composition of the regimens, the beneficial effects (e.g. overall survival or progression free survival) of platinum agents (i.e. cisplatin or carboplatin) in the treatment of ovarian cancer have been demonstrated, e.g. [25,28]. In its technology appraisal 91, the National Institute for Health and Clinical Excellence (NICE) concludes that there is a significant increase in both the medium survival and medium progression free survival, when the patients undergo either a platinum (cisplatin or carboplatin) monotherapy or a paclitaxel-platinum combination therapy (as a first-line chemotherapy) [29]. Moreover, in the same technology appraisal the combination of a platinum-based compound with paclitaxel has been proposed as the recommended option for second-line treatment of women with (partially) platinum-sensitive advanced ovarian cancer. Furthermore, the “Bristol-Myers Squibb” model estimates that the use of a paclitaxel-platinum treatment could result in a gain of 1.73 life-years, compared to a gain of 1.27 years for single-agent paclitaxel [30]. We were unable to find any events related to the administration of a platinum agent in the context of a chemotherapy treatment for ovarian cancer. Neither did the event log contain the trade names “Platinol” (for cisplatin) and/or “Paraplatin” (for carboplatin).

Secondly, the in-depth investigation of the activity sequences of the patients with ovarian cancer that received a paclitaxel based chemotherapy uncovered two important deviations. The activity log did not contain a surgery for 2 out of the 3 patients that were diagnosed with ovarian cancer stage Ia, while this is advised in [23]. In addition, while patients diagnosed with ovarian cancer stage Ic and stage II have an estimated disease recurrence rate between 25-40% and therefore require postoperative paclitaxel chemotherapy, we did not find any patient that received this type chemotherapy.

Additionally, the NICE Guidance TA108 estimates the cost per patient at approximately £4000 for an average of four cycles of treatment. Based on the assumption that one case describes the full treatment of an individual person, we conclude that this event log has an average of approximately 7 treatment cycles and conclude that for only 30% of the patients 4 cycles or less was adequate, see Figure 3. Further research must be performed in order to determine what caused the high deviation of the real number of treatment cycles compared to the prescribed number of treatment cycles.

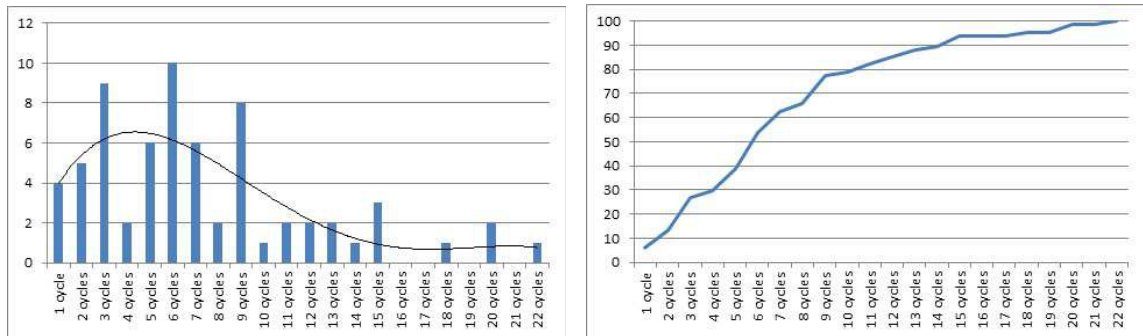


Figure 3: Distribution and cumulative distribution of the number of treatment cycles per patient

4.3. Limitations

The limitations of the current study can be grouped in two categories: the limitations of the constructed event log and the local hospital conditions. The event log inherently suffered from the following limitations: a limited accuracy of the timestamps making performance analyses irrelevant and coarsely recorded additional event information (e.g. the originator information consisted of high-level departmental information). Moreover, we cannot guarantee that the log with a limited time frame covers all the activities of all the patients. The data is retrieved from a financial information system and consequently might not cover all the (non-invoiced) activities in the actual process. Secondly, the event log that was used in this care process analysis study was supplied by a university hospital, which is an important local condition as the log may contain a disproportionate amount of special and/or high risk cases. These limitations, however, do not affect the potential of applying process mining techniques for the continuous quality and performance monitoring of the care processes.

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